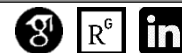


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RESEARCH INTERESTS

- Atmosphere-Ocean Interaction, Tropical Cyclones, Remote Sensing, Upper Atmospheric Physics.

PUBLICATIONS

- Journals: 3
- Preprints: 1
- Conferences: 7

AWARDS & ACCOLADES

- Qualified CSIR-JRF NET – 2019 and GATE GG – 2025.
- DST-SERB International Travel Grant to attend AOGS 2024 in South Korea.
- INSPIRE Scholarship for Higher Education (2015- 20) - DST, Govt. Of India.
- Pathani Samanta Mathematics Scholarship (2012-15) & Medhabruti Scholarship (2015) - Govt. of Odisha.
- Teaching Assistant for the course “Introduction to Computer Sciences CS1101” for 1st Year undergraduate students at IISER Kolkata.

EDUCATION

- Indian Institute of Technology, Bhubaneswar 2020-2025
Ph.D. in Atmosphere and Ocean Sciences, 9.53 CGPA
Courses – Scientific and Technical Writing, Research Methodology and Data Analysis, Physics of Atmosphere and Ocean, Satellite Oceanography and Meteorology, Weather Analysis and Forecasting.
- Indian Institute of Science Education and Research, Kolkata 2015-2020
5 Years BS – MS Dual Degree in Geological Sciences, 8.52 CGPA
Courses – Biogeochemical Cycles, Groundwater Hydrology, Earth System Processes, Remote Sensing and GIS, Atmospheric Sciences, Inverse Theory, Seismology, Mineralogy, Petrology, Palaeontology, Geochemistry.
- Banki Junior College, Banki, Cuttack 2013-2015
12th, Council of Higher Secondary Education, Odisha, 81.83%
- Sri Aurobindo Centre of Integral Education, Banki, Cuttack 2013
10th, Board of Secondary Education, Odisha, 93%

SKILL HIGHLIGHTS

- Skilled in Python, MATLAB, Shell scripting, R, Ferret, HTML, CSS, Web Designing.
- Proficient in remote sensing data analysis, satellite observation datasets, ground-based atmospheric observations, and GLD360, WWLLN lightning datasets.

EXPERIENCE

- Atmosphere – Ocean Interaction (Ongoing)
Ph.D. - IIT Bhubaneswar 24th August 2020 - ongoing
Title: “Investigation of the Role of Geophysical Parameters on Tropical Cyclone Activity in Recent Years”
Supervisor: Dr. D. Swain, Assoc. Professor, SEOCS, Indian Institute of Technology Bhubaneswar, IN
- ESCS Fani: A Study of its Meteorological & Electrical Properties and its Role in Coupling of the Lower and Upper Atmosphere (Completed)
Master’s Thesis - KSKGRL Prayagraj August 2019 - April 2020
Title: “Extremely Severe Cyclonic Storm Fani: A Study of Its Meteorological and Electrical Properties and Its Role in Coupling the Lower and Upper Atmosphere”
Supervisor: Dr. Rajesh Singh, Professor F, KSKGRL, Indian Institute of Geomagnetism, Prayagraj, IN

- Properties of Mesoscale Convective System (MCS), producing TLEs in India (Completed)
Summer Project – KSKGRL Prayagraj May 2018 - July, 2018

I was interested in the occurrences of electrical discharges over the cloud systems and sometimes reaching the base of Ionosphere, known as Transient Luminous Events or TLEs. I characterized and studied 10 such TLEs (9 Sprites and 1 Gigantic Jet) over the central Indian region along with their parent thunderstorm properties. The findings of this study are anticipated to be published in a scientific journal. I carried out this study under the guidance of Prof. Rajesh Singh at K S Krishnan Geomagnetic Research Laboratory, Indian Institute of Geomagnetism, Prayagraj, IN.

- Atmospheric Electrical Discharges (Completed)

Summer Project – KSKGRL Prayagraj

May, 2017 - July, 2017

I studied 3 TLEs (Sprites) and tried to study the relationship between such high-altitude electrical phenomena with low-level thunderstorms. I carried out this study under the guidance of Prof. Rajesh Singh at K S Krishnan Geomagnetic Research Laboratory, Indian Institute of Geomagnetism, Prayagraj, IN.

PUBLICATIONS (Journals)

1. Ray, S, **Moharana, SS**, Nag, A, & Swain, D. (2025). Evaluation of Oceansat-3 gridded wind accuracy for low to extreme wind speed conditions. *Int J Remote Sens*, 46(17), 6403–6427. <https://doi.org/10.1080/01431161.2025.2538827>
2. Patil, OM, **Moharana, SS**, Maurya, AK, Parihar, N, Singh, R, & Dimri, AP. (2024). Role of lightning activity in deciphering atmospheric gravity waves (AGWs) induced D-region ionospheric perturbations during extremely severe cyclonic storm (ESCS) Fani. *J Geophys Res: Space Phys*, 129, e2023JA032187. <https://doi.org/10.1029/2023JA032187>
3. **Moharana, SS**, & Swain, D. (2023). On the recent increase in Atlantic Ocean hurricane activity and influencing factors. *Nat Hazards*, 118, 1387-1399. <https://doi.org/10.1007/s11069-023-06061-3>
4. Ray, S, **Moharana, SS**, & Swain, D. (2022). Ocean response to Tropical Cyclone “Asani”: Surface cooling restricted by Coastal Downwelling. [Preprint] <https://doi.org/10.21203/rs.3.rs-2001209/v1>

CONFERENCES

1. Moharana, S. S. & Swain, D. (2024). Tropical Cyclone Response to Recent Warming in Arabian Sea. Abstract ID: AS11-A001. In the *21st Annual Meeting of the Asia Oceania Geosciences Society 2024 (AOGS2024)*, Pyeongchang, Gangwon-do, South Korea. 23-28 June 2024.
2. Moharana, SS & Swain, D. (2024). Impact of Geophysical Parameters on Tropical Cyclogenesis Locations in the Bay of Bengal. *WOSC 2024*, 27-29 February 2024.
3. Moharana, S. S. & Swain, D. (2023). Cluster Analysis of Tropical Cyclogenesis Locations in the Bay of Bengal. Abstract ID: ABS-05-0004. In the *8th National Conference of Ocean Society of India 2023 (OSICON-23)*, INCOIS, Hyderabad, India. 23-25 August 2023.
4. Moharana, SS & Swain, D. (2022). Investigation of the Impact of Geophysical Parameters Modulating the Recent Rise in Hurricane Activity in the Atlantic Ocean. In *Fall Meeting 2022. AGU*, 1106035.
5. Moharana, SS & Swain, D. (2022). Recent changes in the trends of Atlantic Hurricanes and the factors influencing them. *AOGS 2022, Virtual*, AS11/A022.
6. Moharana, SS & Swain, D. (2021). Atlantic hurricanes of 2020 and the factors influencing them. *OSICON-21*, Online, 12-14 August 2021, OSICON2021/155.
7. Moharana, SS & Singh, R. (2020). Probing a post monsoon Mesoscale Convective System (MCS) and the generated Transient Luminous Events (TLEs) over Indian region. *EGU General Assembly 2020*, Online, 4–8 May 2020, EGU2020-19407. <https://doi.org/10.5194/egusphere-egu2020-19407>.

PERSONAL INFO

- Date of Birth : 20th March, 1998
- Languages Spoken : Odia, English, Hindi, Bengali



Place: Jatani, India
Date: 25th July, 2026